



Report 2

Network Design Using Cisco Packet Tracer
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**NETWORK SECURITY –
BATCH 2**



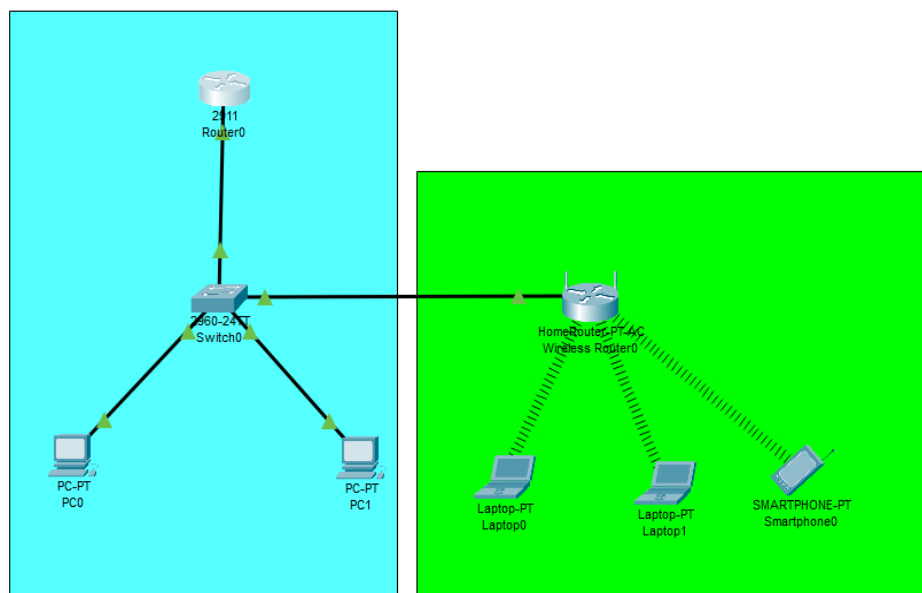
Objective:

To design and configure a small network using a router, a switch, two PCs connected through Ethernet cables, and two laptops with one smartphone connected wirelessly through a Home Wireless Router (Access Point). The objective is to verify network connectivity by successfully communicating between all devices.

Devices Used:

- 1 Cisco Router
- 1 Cisco 2960 Switch
- 1 Home Wireless Router (Access Point)
- 2 Desktop PCs
- 2 Laptops
- 1 Smartphone
- Copper Straight-Through Ethernet Cables

Network Topology:



IP Addressing Scheme:

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.1.1	255.255.255.0	N/A
PC 0	192.168.1.10	255.255.255.0	192.168.1.1
PC 1	192.168.1.11	255.255.255.0	192.168.1.1
Laptop 0	192.168.1.20	255.255.255.0	192.168.1.1
Laptop 1	192.168.1.21	255.255.255.0	192.168.1.1
Smartphone	192.168.1.30	255.255.255.0	192.168.1.1

Configuration:

Router 0

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int e0/0
%Invalid interface type and number
Router(config)#int g0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1    YES manual up          down
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/3/0              unassigned      YES unset  administratively down down
Serial0/3/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router#show arp

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

PC 0 Config:

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:2FFF:FE8B:28EE

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

PC 1 Config:

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2D0:FFFF:FE82:AC04

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Laptop 0:

Laptop0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: Wireless0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.20

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address: /

Link Local Address: FE80::201:43FF:FE20:E4CB

Default Gateway:

DNS Server:

Laptop 1:

Laptop1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: Wireless0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.21

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address: /

Link Local Address: FE80::201:42FF:FE3C:B58

Default Gateway:

DNS Server:

Smart Phone

Smartphone0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: Wireless0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.30

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static IPv6 request failed.

IPv6 Address: /

Link Local Address: FE80::207:ECFF:FE10:A686

Default Gateway:

DNS Server:

Wireless Router:

Wireless Router0

Physical **Config** GUI Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless 2.4G

Wireless 5G(1)

Wireless 5G(2)

Wireless Guest 2.4G

Wireless Guest 5G(1)

Wireless Guest 5G(2)

LAN Settings

IP Configuration

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

PC 0 to Phone Ping:

```
C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Reply from 192.168.1.30: bytes=32 time=47ms TTL=128
Reply from 192.168.1.30: bytes=32 time=19ms TTL=128
Reply from 192.168.1.30: bytes=32 time=22ms TTL=128
Reply from 192.168.1.30: bytes=32 time=20ms TTL=128

Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 47ms, Average = 27ms

C:\>|
```

Laptop To PC 0:

```
C:\>
C:\>
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=28ms TTL=128
Reply from 192.168.1.10: bytes=32 time=21ms TTL=128
Reply from 192.168.1.10: bytes=32 time=24ms TTL=128
Reply from 192.168.1.10: bytes=32 time=22ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 28ms, Average = 23ms

C:\>|
```

PC 0 To Laptop 2:

```
C:\>ping 192.168.1.21

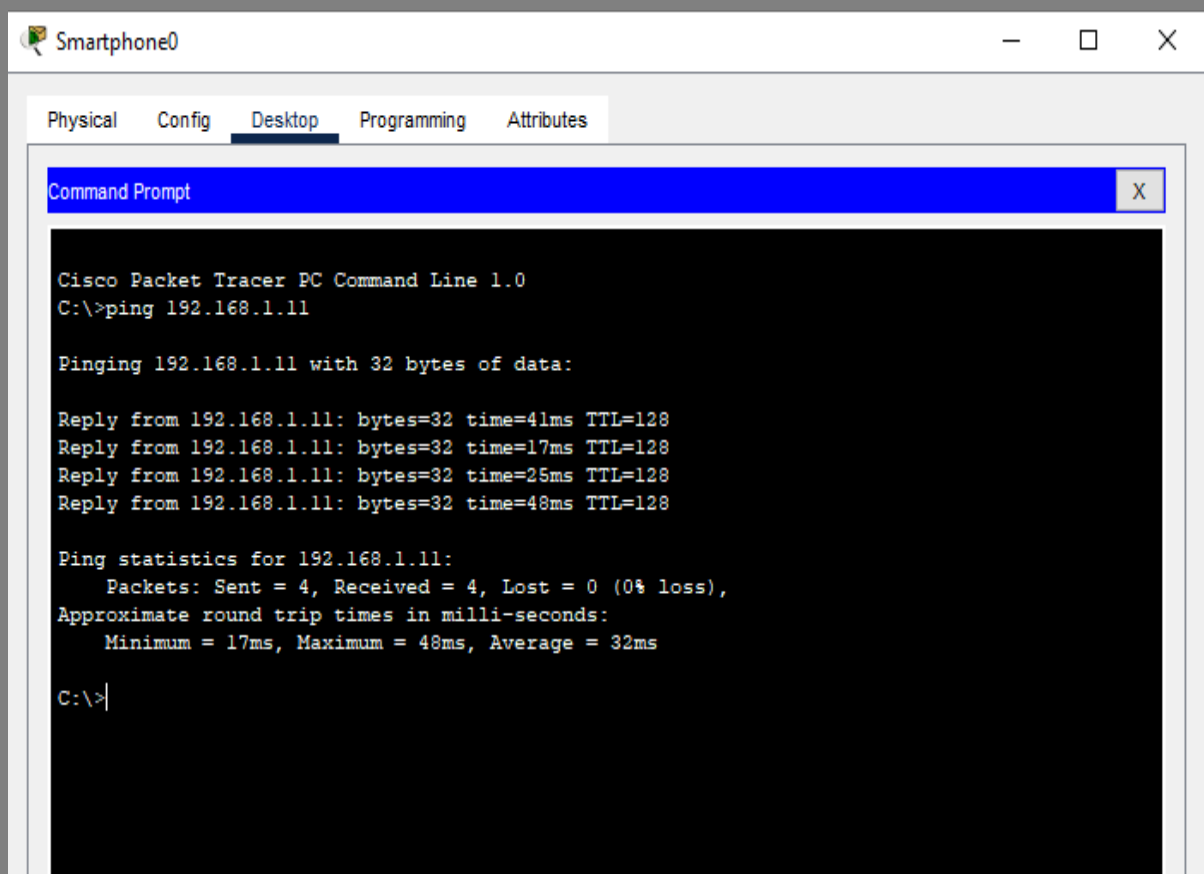
Pinging 192.168.1.21 with 32 bytes of data:

Reply from 192.168.1.21: bytes=32 time=39ms TTL=128
Reply from 192.168.1.21: bytes=32 time=16ms TTL=128
Reply from 192.168.1.21: bytes=32 time=14ms TTL=128
Reply from 192.168.1.21: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.1.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 39ms, Average = 21ms

C:\>
```

Phone To PC 2:



Result:

The small wired and wireless network was successfully designed and configured in Cisco Packet Tracer. All wired and wireless devices were able to communicate with each other successfully, confirming that the network configuration was correct.

Conclusion:

This lab provided practical experience in creating a basic home/office network using Cisco Packet Tracer. It demonstrated how wired and wireless devices can be connected through a router, switch, and access point. The successful ping test verified proper IP addressing, connectivity, and communication between all devices.

Task 2: RJ45 Ethernet Cable (T568B Standard)

Objective:

To prepare a straight-through Ethernet cable using the **T568B wiring standard** by attaching RJ45 connectors on both ends of a UTP cable. The cable will be used to connect networking devices such as a PC to a switch or a router to a switch.

Materials Required:

- Cat5e/Cat6 UTP Cable
- 2 × RJ45 Connectors
- RJ45 Crimping Tool
- Wire Stripper/Cutter

T568B Wiring Standard:

Pin	Wire Colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Procedure:

- Cut the UTP cable to the required length.
- Strip approximately 2–3 cm of the outer jacket from both ends of the cable.
- Untwist the four wire pairs and straighten all eight wires.
- Arrange the wires according to the T568B standard in the correct order.
- Trim the wires evenly so they are the same length.
- Hold the RJ45 connector with the clip facing downward and insert the wires fully into the connector.
- Ensure that all eight wires reach the front of the connector and that the cable jacket is inside the connector.
- Use the RJ45 crimping **tool** to crimp the connector securely.
- Repeat the same T568B wiring sequence on the other end of the cable.
- Test the completed cable using a LAN cable tester to verify proper connectivity.

Cable Wiring:

RJ45 Connectors



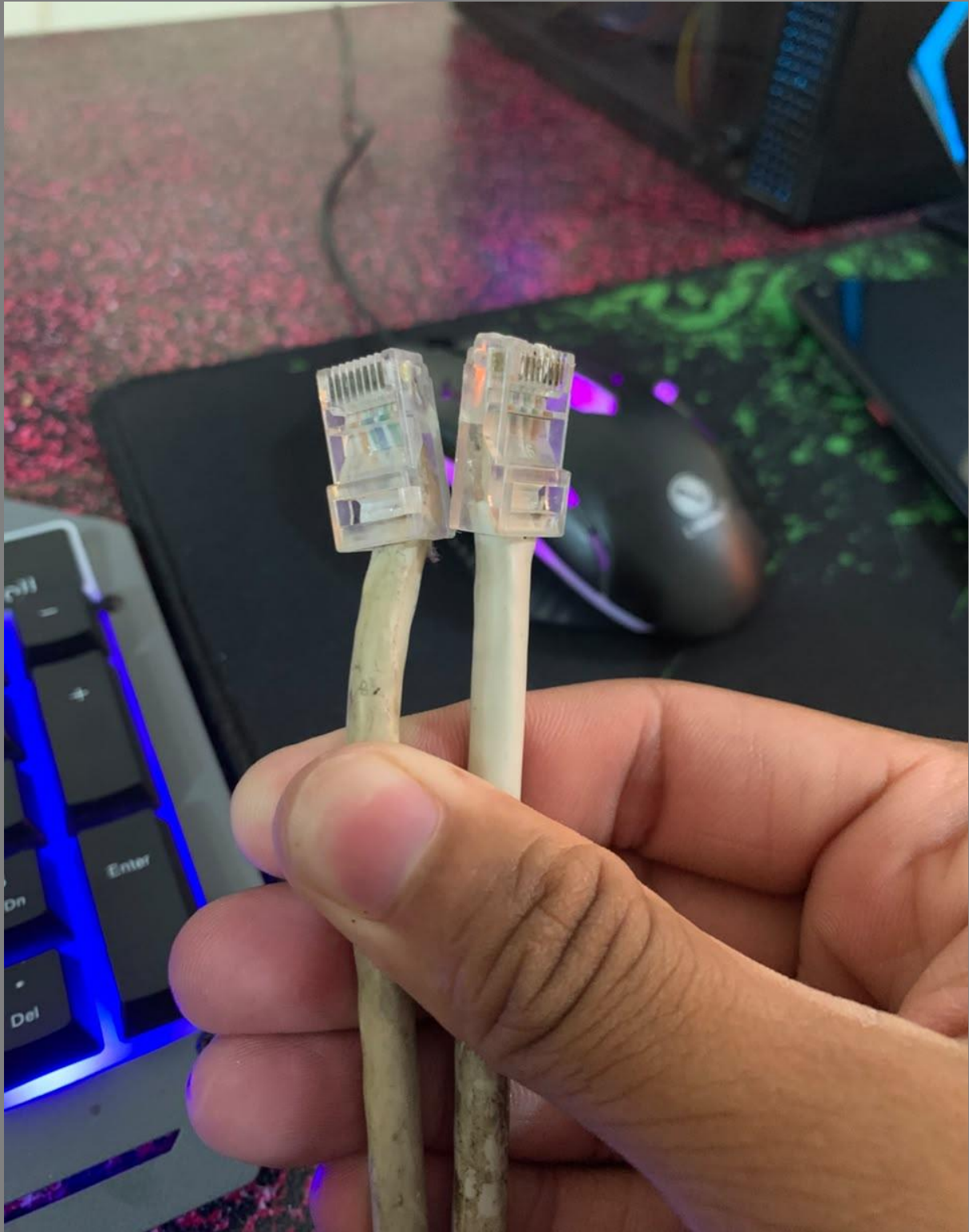
Trim the wires evenly so they are the same length



Arrange the wires according to the T568B standard in the correct order



Hold the RJ45 connector with the clip facing downward and insert the wires fully into the connector. Ensure that all eight wires reach the front of the connector and that the cable jacket is inside the connector. Use the RJ45 crimping tool to crimp the connector securely. Repeat the same T568B wiring sequence on the other end of the cable.



Result:

The straight-through Ethernet cable was successfully prepared using the T568B standard. The cable passed the continuity test, confirming that all eight wires were correctly connected and working properly.

Conclusion:

This lab provided practical experience in creating a standard T568B straight-through Ethernet cable. It helped in understanding the correct wire sequence, the use of RJ45 connectors, and the cable crimping process. The completed cable can be used to connect networking devices such as a computer to a switch or a router to a switch.